

Case Study: Big Data Security Issues at "DataSphere Corporation"

Introduction:

DataSphere Corporation is a global leader in data analytics, serving a wide range of industries from healthcare to finance. With the advent of big data technologies, DataSphere has been able to process and analyze vast amounts of data to provide valuable insights to its clients. However, the rapid growth in data volume and complexity has introduced significant security challenges.

Scenario:

DataSphere recently migrated its infrastructure to a cloud-based big data platform to enhance scalability and reduce costs. The platform supports multi-tenancy, allowing multiple clients to share the same infrastructure. However, this move has exposed DataSphere to a range of security threats.

The Incident:

One morning, the IT team at DataSphere detected an unusual spike in network traffic. Upon investigation, they discovered that they were under a Distributed Denial of Service (DDoS) attack. The attackers flooded the network with excessive requests, overwhelming the servers and causing a major outage. This impacted the availability of services, and many clients were unable to access their data.

Tasks:

- A. Design a scenario for DDoS attack on DataSphere's infrastructure.
- B. Analyze the impact of this attack on DataSphere's infrastructure.
- C. Determine the extent of the data breach and identify the affected clients.
- D. Implement enhanced security measures to protect sensitive client data from unauthorized access.